

LAB EXPERIMENT – 3

Building a Shared Repository with DagsHub and MLflow for Collaborative MLOps

Course Outcome: CO1

Aim

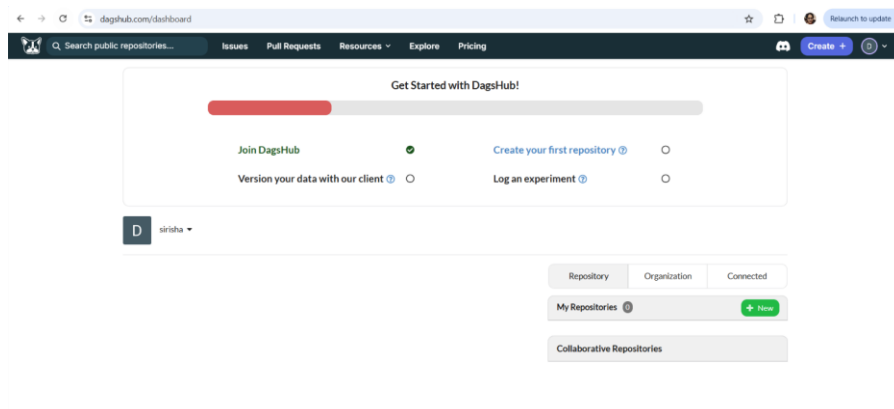
To create a collaborative machine learning repository using **DagsHub** and track experiments using **MLflow** in **Google Colab**.

Steps

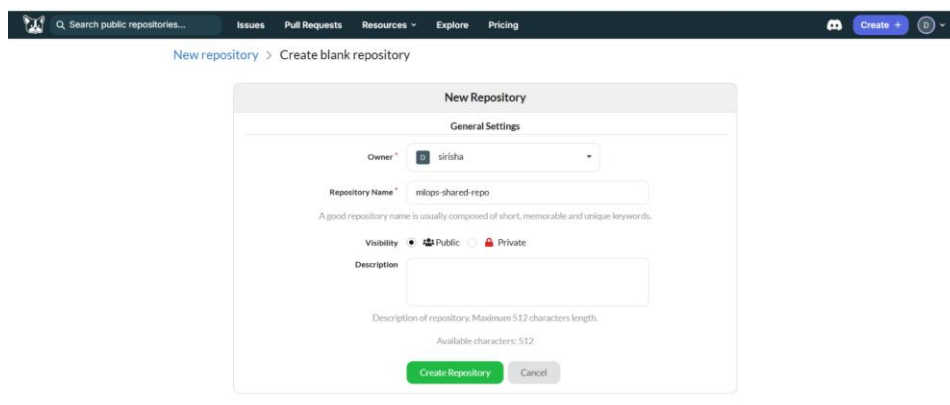
1. Create a shared repository in DagsHub.
2. Connect Google Colab with Git and DagsHub.
3. Train a simple ML model.
4. Track parameters, metrics, and artifacts using MLflow.
5. Push experiment files to the shared DagsHub repository.

Step 1: Create a DagsHub Repository

1. Open <https://dagshub.com>
2. Sign in.



3. Click **New Repository**.



4. Repository name: mlops-shared-repo

5. Choose **Public**.
6. Click **Create Repository**.

Copy the repository URL, for example:

<https://dagshub.com/username/mlops-shared-repo.git>

Ex:

<https://dagshub.com/sirisha/mlops-shared-repo.git>

Step 2: Open Google Colab

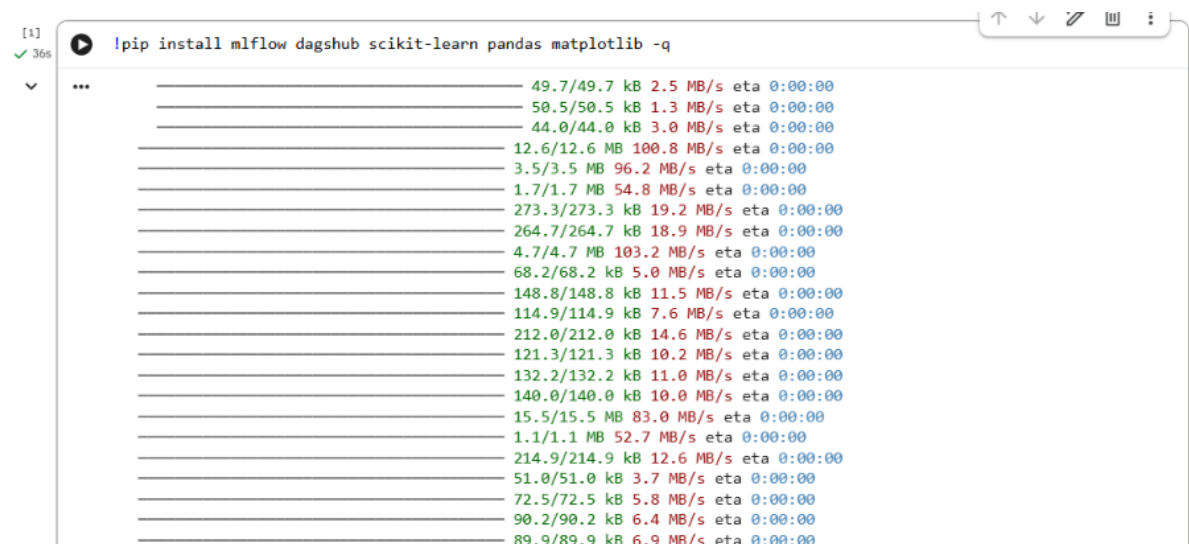
Create a new notebook.

Skill-3 MLOPs.ipynb

Step 3: Install Required Packages

Run the following cell:

`!pip install mlflow dagshub scikit-learn pandas matplotlib -q`



```
[1] 36s !pip install mlflow dagshub scikit-learn pandas matplotlib -q
***
49.7/49.7 kB 2.5 MB/s eta 0:00:00
50.5/50.5 kB 1.3 MB/s eta 0:00:00
44.0/44.0 kB 3.0 MB/s eta 0:00:00
12.6/12.6 MB 100.8 MB/s eta 0:00:00
3.5/3.5 MB 96.2 MB/s eta 0:00:00
1.7/1.7 MB 54.8 MB/s eta 0:00:00
273.3/273.3 kB 19.2 MB/s eta 0:00:00
264.7/264.7 kB 18.9 MB/s eta 0:00:00
4.7/4.7 MB 103.2 MB/s eta 0:00:00
68.2/68.2 kB 5.0 MB/s eta 0:00:00
148.8/148.8 kB 11.5 MB/s eta 0:00:00
114.9/114.9 kB 7.6 MB/s eta 0:00:00
212.0/212.0 kB 14.6 MB/s eta 0:00:00
121.3/121.3 kB 10.2 MB/s eta 0:00:00
132.2/132.2 kB 11.0 MB/s eta 0:00:00
140.0/140.0 kB 10.0 MB/s eta 0:00:00
15.5/15.5 MB 83.0 MB/s eta 0:00:00
1.1/1.1 MB 52.7 MB/s eta 0:00:00
214.9/214.9 kB 12.6 MB/s eta 0:00:00
51.0/51.0 kB 3.7 MB/s eta 0:00:00
72.5/72.5 kB 5.8 MB/s eta 0:00:00
90.2/90.2 kB 6.4 MB/s eta 0:00:00
89.9/89.9 kB 6.9 MB/s eta 0:00:00
```

Step 4: Configure Git in Colab

Replace with your details.

`!git config --global user.name "Your Name"`

`!git config --global user.email yourmail@gmail.com`

Verify:

`!git config --global --list`

```
[2] |git config --global user.name "Sirisha-Potluri2011"
    |git config --global user.email sirisha@klh.edu.in

[3] |git config --global --list
    user.name=Sirisha-Potluri2011
    user.email=sirisha@klh.edu.in
```

Step 5: Clone the DagsHub Repository

Generate a DagsHub access token

- Open <https://dagshub.com> and log in.
- Click your profile picture → Settings.
- Open Access Tokens.
- On the DagsHub page, click the default token entry.
- Copy the token string that appears.
- Go back to your Colab notebook.
- Create a new code cell and run the following code.

TOKEN = "PASTE_YOUR_TOKEN_HERE"

```
!git clone https://$TOKEN@dagshub.com/sirisha/mlops-shared-repo.git
```

```
%cd mlops-shared-repo
```

```
!ls -la
```

```
TOKEN = "9027e50ebbbcb06ad48acc0df8ef10c81f1fb172"

!git clone https://$TOKEN@dagshub.com/sirisha/mlops-shared-repo.git

%cd mlops-shared-repo

!ls -la

... Cloning into 'mlops-shared-repo'...
warning: You appear to have cloned an empty repository.
/content/mlops-shared-repo
total 12
drwxr-xr-x 3 root root 4096 Jul 31 04:14 .
drwxr-xr-x 1 root root 4096 Jul 31 04:14 ..
drwxr-xr-x 7 root root 4096 Jul 31 04:14 .git
```

Step 6: Create a Sample Dataset and Train a Model

```
from sklearn.datasets import load_iris
```

```
from sklearn.model_selection import train_test_split
```

```
from sklearn.ensemble import RandomForestClassifier
```

```
from sklearn.metrics import accuracy_score
```

```
import pandas as pd
```

```

iris = load_iris()

X_train, X_test, y_train, y_test = train_test_split(
    iris.data, iris.target, test_size=0.2, random_state=42
)

model = RandomForestClassifier(n_estimators=100, random_state=42)

model.fit(X_train, y_train)

pred = model.predict(X_test)

acc = accuracy_score(y_test, pred)

print("Accuracy:", acc)

```

Expected output:

Accuracy: 1.0



```

from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score
import pandas as pd

iris = load_iris()
X_train, X_test, y_train, y_test = train_test_split(
    iris.data, iris.target, test_size=0.2, random_state=42
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model = RandomForestClassifier(n_estimators=100, random_state=42)
model.fit(X_train, y_train)
pred = model.predict(X_test)
acc = accuracy_score(y_test, pred)
print("Accuracy:", acc)

```

... Accuracy: 1.0

Step 7: Configure MLflow with DagsHub

Generate DagsHub Token

1. DagsHub → Settings → Access Tokens → Generate Token.
2. Copy the token.

Run:

```

import dagshub

import mlflow

dagshub.init(
    repo_owner="username",
    repo_name="mlops-shared-repo",
    mlflow=True
)

```

When prompted, authorize your account.

```
import dagshub
import mlflow
dagshub.init(
    repo_owner="sirisha",
    repo_name="mlops-shared-repo",
    mlflow=True
)

...

!!! AUTHORIZATION REQUIRED !!!

Open the following link in your browser to authorize the client:
https://dagshub.com/login/oauth/authorize?state=ea879452-2d9d-4306-a3b8-49098068961e&client\_id=32b60ba385aa7cecf24046d8195a71c07dd345d9

Accessing as sirisha
Initialized MLflow to track repo "sirisha/mlops-shared-repo"
Repository sirisha/mlops-shared-repo initialized!
```

Step 8: Track Experiment with MLflow

with mlflow.start_run():

```
mlflow.log_param("n_estimators", 100)
```

```
mlflow.log_metric("accuracy", acc)
```

```
mlflow.sklearn.log_model(model, "random_forest_model")
```

with open("result.txt", "w") as f:

```
f.write(f"Accuracy: {acc}")
```

```
mlflow.log_artifact("result.txt")
```

```
print("Experiment logged successfully")
```

```
with mlflow.start_run():
    mlflow.log_param("n_estimators", 100)
    mlflow.log_metric("accuracy", acc)
    mlflow.sklearn.log_model(model, "random_forest_model")
    with open("result.txt", "w") as f:
        f.write(f"Accuracy: {acc}")
    mlflow.log_artifact("result.txt")
    print("Experiment logged successfully")

... 2026/07/31 04:22:25 WARNING mlflow.models.model: `artifact_path` is deprecated. Please use `name` instead.
★ View run masked-fawn-215 at: https://dagshub.com/sirisha/mlops-shared-repo/mlflow/#/experiments/0/runs/262521e5d58b45f2a7599ad643871
🌱 View experiment at: https://dagshub.com/sirisha/mlops-shared-repo/mlflow/#/experiments/0
Experiment logged successfully
```

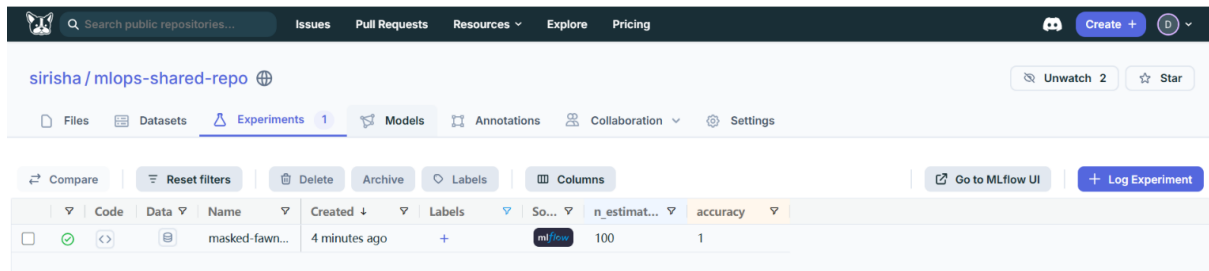
Step 9: Verify MLflow Run

Run:

```
print(mlflow.get_tracking_uri())
```

```
[9] ✓ Os print(mlflow.get_tracking_uri())
... https://dagshub.com/sirisha/mlops-shared-repo/mlflow
```

Open your DagsHub repository and click **Experiments** to see:



Run this in a new cell:

```
import mlflow

import dagshub

dagshub.init(
    repo_owner="sirisha",
    repo_name="mlops-shared-repo",
    mlflow=True
)

mlflow.set_experiment("MLOps_Skill_3")

with mlflow.start_run():
    mlflow.log_param("model", "RandomForest")
    mlflow.log_param("n_estimators", 100)
    mlflow.log_metric("accuracy", 0.95)
    mlflow.log_metric("f1_score", 0.94)
    mlflow.sklearn.log_model(model, "model")
```

Open your DagsHub repository and click **Experiments** to see:

- Parameters
- Metrics
- Model artifact
- Result file

	Code	Data	Name	Created	Labels	So...	n_estimat...	model	accuracy	f1_score
☐			nervous-kite-...	a minute ago	+		100	RandomForest	0.95	0.94
☐			masked-fawn...	7 minutes ago	+		100		1	

Step 10: Save Notebook Outputs

`!mkdir outputs`

`!cp result.txt outputs/`

```
!mkdir outputs
!cp result.txt outputs/
```

Step 11: Commit Files to DagsHub Repository

`!git add .`

`!git commit -m "Added MLflow experiment and outputs"`

```
[12]
✓ Os !git add .
!git commit -m "Added MLflow experiment and outputs"

... [main (root-commit) 0eac30c] Added MLflow experiment and outputs
2 files changed, 2 insertions(+)
create mode 100644 outputs/result.txt
create mode 100644 result.txt
```

Step 12: Push Changes to DagsHub

Use username and token authentication.

`!git branch`

```
[13]
✓ Os !git branch

... * main
```

`!git push origin main`

If your default branch is master, use:

`!git push origin master`

```
[14]
✓ Is !git push origin main

... Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (4/4), 335 bytes | 335.00 KiB/s, done.
Total 4 (delta 0), reused 0 (delta 0), pack-reused 0
To https://dagshub.com/sirisha/mlops-shared-repo.git
* [new branch] main -> main
```

Step 13: Verify Collaboration

Open the DagsHub repository and confirm:

- Notebook files
- Output files
- MLflow experiments
- Model artifacts

